1.5 Exploring Rational Exponents

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<tr><td>Benchmark</td><td colspan="2">MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.</td><td>Learning Target</td><td>• I can apply the Laws of Exponents to rational exponents.
• I can convert between expressions involving rational exponents and expressions involving radicals.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">• How can a number with a rational number exponent represent the same value in radical form?
• What is the relationship between a number with a rational exponent and a number in radical form?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">In this lesson, students will continue their learning with the Laws of Exponents to focus on generating equivalent numerical expressions. Students will explore the relationship between rational number exponents and radical form. They will practice how to rewrite expressions between forms and evaluate them.</td></tr>
<tr><td rowspan="4">Component Summary</td><td>1.5.1 Warm-Up (~5 min.)</td><td>Bell Work on the Laws of Exponents (SE p. 27)</td><td>1.5.4 Exploration (5 min.)</td><td>Explore roots other than square and cube with translating between rational exponent and radical expression (SE p. 30)</td></tr>
<tr><td>1.5.2 Exploration (10 min.)</td><td>Explore generating equivalent expressions and evaluating rational exponents (SE p. 28)</td><td>1.5.5 Guided Instruction (10 min.)</td><td>Instruction on applying exponent laws with rational exponents (SE p. 30)</td></tr>
<tr><td>1.5.3 Guided Instruction (10 min.)</td><td>Instruction on understanding the relationship between rational exponents and radical expressions (SE p. 29)</td><td>1.5.6 Try It (5 min.)</td><td>Practice rewriting expressions using the Laws of Exponents (SE p. 31)</td></tr>
<tr><td>1.5.7 Wrap-Up (~5 min.)</td><td colspan="3">Ticket Out the Door to write an equivalent expression using the Laws of Exponents. (SE p. 32)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td>Materials</td><td>Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms</u>: N/A

<u>Additional Terms:</u>
• Rational Exponent
• Radical Form
• Equivalent Expression</td><td>• Calculator for 1.5.1 Warm-up</td><td>• Work through each of the explorations prior to having students try these. They are designed to lead students to a discovery in a very specific way.</td></tr>
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<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not make the connection between the two forms.
Students may put the numerator of the fraction in the exponent as the index of the radical form.
Students may make calculation errors when determining the value of the expression.</td><td>This concept is new to most students and may be abstract in their thinking. It’s important to avoid any tricks with where values go and focus on the actual mechanics of why the two forms (radical and rational exponent) are equivalent.
If time is an issue, then consider assigning 1.5.6 Try It for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
During 1.5.5 Guided Instruction and 1.5.6 Try It, students are given the opportunity to try a question on their own. Consider leveraging the Think-Pair-Share protocol to allow students to work with a partner after they’ve done it on their own and discuss their methods and findings.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students have previously worked with Laws of Exponents and square and cube roots. This lesson prompts students to apply that knowledge to establish that the Laws of Exponents apply to rational exponents.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Pay attention to student performance and response during the exploration and guided practice activities. Students who are struggling should receive differentiated support during activity 1.5.6 Try It.
Consider creating a visual aid that represents the information discovered during 1.5.3 Guided Instruction. Continually use the language of the re-writing process to help auditory learners remain connected to the content.
The “Wrap-Up” section of the lesson provides a formative assessment that provides data on student mastery of the content. This data could be used to determine future differentiation needs.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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1.5.1 Warm-Up: Bell Work

Component Overview: *Students will complete a Bell Work activity, applying what they learned about Laws of Exponents from the prior lesson to generate equivalent expressions and evaluate to find the value. This activity is designed to be done independently.*

1.5.1 Warm-Up: Bell Work

1. Use the laws of exponents to write an equivalent expression. Then, give the value of each expression. If necessary, use a calculator to find the value of the equivalent expression.

$(8^{-5})(8^3)$ Equivalent Expression $\frac{1}{8^2}$ Value $\frac{1}{64}$	$(3^4)^6$ Equivalent Expression 3^{24} Value 282,429,536,481
$\left(\frac{2}{11}\right)^3$ Equivalent Expression $\frac{2^3}{11^3}$ Value $\frac{8}{1331}$	$\left(\frac{12}{36}\right)^{-4}$ Equivalent Expression $\frac{36^4}{12^4}$ Value 81

For this Warm-Up, consider allowing students to use a calculator to find the value of these expressions. While the first one is appropriate without a calculator, students may spend an excessive amount of time trying to find the value of 3^{24} instead of moving onto the next questions in this activity.

It is possible that students give equivalent expressions that are not in the form of the answer key. Allow the students the freedom of writing an equivalent expression in a variety of ways. If students provide a variety of responses, consider sharing the variety with the class to expand all students' understanding of equivalent expressions.

If time allows, consider asking the following while reviewing the answers:

- *What Law of Exponents did you use to generate the equivalent expression?*
- *How did you know you could use the law you chose?*
- *How did using the Law of Exponents help you determine the value?*

1.5.2 Exploration: Rational Exponents

Component Overview: *Students will work with a partner to conduct an exploration with generating equivalent expressions and evaluating rational exponents. For this activity, begin by releasing students to complete question 1 through 3 with their partner and call the class back together to discuss their answers. Repeat this process for questions 4 and 5, 6 – 9, and 10 – 13.*

1.5.2 Exploration: Rational Exponents

1. Fill in the blank with the value of each.

   $16^0 = 1$ $\qquad$ $16^1 = 16$

What Law of Exponents did you use to determine the value of 16^0?

2. Using the values above, guess the value of $16^{\frac{1}{2}}$.
   Answers will vary. I guess that it would be somewhere between 1 and 16. I guess that it would be 8, because that's half of 16.

3. Explain the strategy you used.
   Answers will vary. I found out half of the value when raised to the power of 1. I decided to use a value that was halfway to 16.

After questions 1 – 3, consider having students share out the different guesses that they came up with for the value in question 2. Poll the class to determine who came up with that same value, charting each guess on the board until all guesses have been accounted for. Have students also share their strategy for determining their guesses to allow other students to hear different approaches to the problem.

4. Find the value of $16^{\frac{1}{2}}$ using a calculator.
   4

5. Discuss with your partner what you notice about the relationship between 16 and the value of $16^{\frac{1}{2}}$.
   Answers will vary. The value is the same as the square root of 16.

After questions 4 and 5, consider using an on-screen calculator, document camera, or other method to show students how to input a fractional exponent into a calculator. Students may struggle with this skill.

6. Predict the value of $49^{\frac{1}{2}}$. Check using your calculator.
   7

7. Predict the value of $100^{\frac{1}{2}}$. Check using your calculator.
   10

8. Write an expression equivalent to $a^{\frac{1}{2}}$.
   $\sqrt{a}$

1.5.2 Exploration: Rational Exponents (continued)

9. Check your answer with your partner. If necessary, update your answer.
   Answers may vary. Students should arrive at correct answer after collaboration.

10. Predict the value of $8^{\frac{1}{3}}$. Check using your calculator.
    2

11. Predict the value of $125^{\frac{1}{3}}$. Check using your calculator.
    5

12. Write an expression equivalent to $a^{\frac{1}{3}}$.
    $\sqrt[3]{a}$

13. Check your answer with your partner. If necessary, update your answer.
    Answers may vary. Students should arrive at correct answer after collaboration.

After questions 6 – 9, engage students in a discussion about what they uncovered in question 8. Have students explain how they chose the answer that they did from the work they did in questions 5 – 7.

- *What relationship did you notice between the values in the rational exponent and the values in the radical form?*
- *Do you think those patterns would hold true for any value outside of 2 and 3?*

1.5.3 Guided Instruction: Why Does That Work?

Component Overview*: Students will receive guided instruction on understanding the relationship between rational exponents and radical expressions.*

1.5.3 Guided Instruction: Why Does That Work?

1. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}}$	$6^{\frac{1}{2}+\frac{1}{2}} = 6^1$	6

2. What is the whole number that completes this equation?

$$\sqrt{6} \cdot \sqrt{6} = 6$$

Because $6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}} = 6$ and $\sqrt{6} \cdot \sqrt{6} = 6$, then $6^{\frac{1}{2}}$ is equivalent to $\sqrt{6}$.

3. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}}$	$5^{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}} = 5^1$	5

4. What is the whole number that completes the equation?

$$\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = 5$$

Because $5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} = 5$ and $\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = 5$, then $5^{\frac{1}{3}}$ is equivalent to $\sqrt[3]{5}$.

5. Rewrite each exponential expression as a radical expression. Then, evaluate each expression.

Exponential Expression	Radical Expression	Value
$81^{\frac{1}{2}}$	$\sqrt{81}$	9
$8^{\frac{1}{3}}$	$\sqrt[3]{8}$	2
$(-216)^{\frac{1}{3}}$	$\sqrt[3]{-216}$	-6

- *What is the relationship between the square root of 6 and 6 raised to the one-half power?*
- *Do you think this relationship would exist with any values or is it strictly for the square roots? Why or why not?*

After questions 1 and 2, engage students in a discussion related to the connection between square roots and values raised to the one-half power. After questions 3 and 4, rekindle the discussion of the relationship between cube roots and values raised to the one-third power. Draw student attention to the process connection between the two different types.

Challenge students to create an additional row, from scratch, to add to the table in question 5. Have them complete all three columns of their new row.

1.5.4 Exploration: Other Roots

***Component Overview**: Students will explore roots other than square and cube with translating between rational exponent and radical expression. For this activity, review the narrative language prior to the table. Model the first row for students and then release students to work with a partner on the remaining rows.*

1.5.4 Exploration: Other Roots

Rewriting exponential expressions as radical expressions does not only apply to square roots and cube roots. Other roots apply these same principals.

Complete the table to explore other rational exponents and roots.

Rational Exponent	Radical Expression	Value
$81^{\frac{1}{4}}$	$\sqrt[4]{81}$	± 3
$256^{\frac{1}{4}}$	$\sqrt[4]{256}$	± 4
$(-32)^{\frac{1}{5}}$	$\sqrt[5]{-32}$	-2
$\left(1{,}000{,}000{,}000^{\frac{1}{9}}\right)$	$\sqrt[9]{1{,}000{,}000{,}000}$	10

Relate what students have done with cube and square roots in earlier activities to situations where the roots are different. Remind students to refer to the work they just completed in the prior activity as a basis for consideration when completing this exploration. When the activity is complete, have students share out their answers and strategies of how they thought through each question.

- *What part of this process was similar to what you did with cube and square roots in the prior activity?*
- *What parts were different?*

Challenge students to create an additional row, from scratch, to add to the table in this question. Have them complete all three columns of their new row.

1.5.5 Guided Instruction: Exponent Laws With Rational Exponents

Component Overview: *Students will receive guided instruction on applying exponent laws with rational exponents.*

1.5.5 Guided Instruction: Exponent Laws with Rational Exponents

1. Kalini's teacher asked her to rewrite $\left((100)^{\frac{1}{2}}\right)^3$ as an equivalent exponential expression. Kalini says that she can use a law of exponents to multiply the powers.

   a. Which law of exponents is Kalini using?
   Power of a Power

   b. Using the laws of exponents, what is an equivalent exponential expression for $\left((100)^{\frac{1}{2}}\right)^3$?
   $100^{\frac{3}{2}}$

2. The expression $27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}}$ can also be rewritten using the laws of exponents.

   a. Rewrite the expression.

   $27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}} = 27^2$

   b. Which exponent law was used?
   Product of Powers

   c. Check your answer with your partner.
   Answers may vary. Students should end with correct answer after collaboration.

   d. Evaluate the exponential expression.
   729

During the modelling of these questions, allow for students to share what they think about each part of the question, referring to the context. Model for students a way of thinking through each part of the question, using appropriate mathematical language to help reinforce what is happening at each step.

Consider writing each step of the process for questions such as 1b and 2a. For each step, use a different colored marker to label what action was taken at each step in a proof-like fashion. This way, students will be able to visualize each step and have a reference for future questions of a similar nature.

For a challenge, consider having students create a problem using a different rational exponent being rewritten by a different Law of Exponents.

3. Maeko explained his work for rewriting the following expression.

Explanation	$16^{-\frac{8}{2}} \cdot 16^{\frac{1}{2}}$
I first used the product of powers to rewrite the expression.	$16^{-\frac{8}{2} \cdot \frac{1}{2}} = 16^{-\frac{8}{4}}$
Next, I simplified the fractional exponent.	4^{-2}
Finally, I rewrote the expression using the negative exponent law.	$\frac{1}{16}$

Check Maeko's work and explain any errors he may have made.

Maeko made mistakes in the first and second step. In the first step, he multiplied the exponents instead of adding them. When he rewrote the second step, he changed the base to 4.

Think-Pair-Share
Have students complete this question on their own first. Then, allow each student to compare their work with a partner to discuss their approach and final answer. Together, each pair should come to a consensus on the correct answer. Have pairs share out their answer and determine what other pairs came to the same conclusion, repeating until all answers have been heard.

- *What was Maeko attempting to do when he made the mistake? What went wrong in his process?*
- *What advice would you give him to help explain what went wrong and how he can avoid making that mistake again in the future?*

1.5.6 Try It: Writing Rational Exponents

Component Overview: *Students will work independently to practice rewriting expressions using the Laws of Exponents then evaluate the expression.*

1.5.6 Try It: Writing Rational Exponents

Rewrite each expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$5^{\frac{26}{7}} \cdot 5^{\frac{9}{7}}$	5^5	3,125
$\dfrac{23^{\frac{8}{3}}}{23^{\frac{2}{3}}}$	23^2	529
$125^{\frac{7}{15}} \cdot 125^{-\frac{2}{15}}$	$125^{\frac{1}{3}}$	5
$169^{-\frac{5}{22}} \cdot 169^{\frac{25}{22}} \cdot 169^{-\frac{31}{22}}$	$169^{-\frac{1}{2}}$	$\frac{1}{13}$

Challenge students to create an additional row, from scratch, to add to the table in question 5. Have them complete all three columns of their new row.

1.5.7 Wrap-Up: Ticket Out the Door

***Component Overview**: Students will complete a Ticket Out the Door that assesses students' ability to write an equivalent expression using the Laws of Exponents. This activity should be done independently to allow for accurate data on student mastery of the content of this lesson. Students should not use a calculator to complete the Wrap-Up.*

1.5.7 Wrap-Up: Ticket Out the Door

1. Find the value of the variable.

   a. $\sqrt{14} = 14^{\frac{1}{k}}$

   $k = 2$

   b. $50^{\frac{1}{3}} = \sqrt[n]{50}$

   $n = 3$

2. Complete the table by writing an equivalent expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$6^{-\frac{1}{6}} \cdot 6^{\frac{19}{6}}$	6^3	216
$\dfrac{7^{\frac{1}{2}}}{7^{\frac{5}{2}}}$	7^{-2}	$\dfrac{1}{49}$

Student performance on this Wrap-Up can provide data that indicates additional support and differentiation that may be necessary. The following can be determined:

- *Question 1*
  - *Part A: Rewriting radical form to rational exponent form*
  - *Part B: Rewriting rational exponent form to radical form*
- *Question 2*
  - *Equivalent Expression*
  - *Calculating Actual Value*